Haoran Geng

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U.S. Permanent Resident

Research Interests

Computer Architecture; Memory Systems; Processing-in-Memory (PIM); Compute-in-Memory (CIM); Hardware Security and Secure Memory; Heterogeneous Computing; Hardware-Software Co-Design; Architecture-Aware Performance Modeling and Design Automation.

Education

University of Notre Dame, Notre Dame, IN *2021 – Present*
Ph.D., Computer Science and Engineering (Expected 2026)
Advisors: Prof. Xiaobo Sharon Hu and Prof. Michael Niemier

Columbia University, New York, NY *2018 – 2021*
M.S., Computer Engineering

Rose-Hulman Institute of Technology, Terre Haute, IN *2014 – 2018*
B.S., Computer Engineering

Research Experience

Graduate Research Assistant, University of Notre Dame *Feb 2021 – Present*

- Conducting research in computer architecture, memory systems, processing-in-memory, and hardware security.
- Leading **COSMOS**, an RL-based counter-cache optimization project for secure memory; published as first author at MICRO 2025.
- Developing memory-centric and in-memory computing architectures for privacy-preserving computation, encryption, and emerging cryptographic workloads.
- Developing **VIPER**, an architecture-aware analytical framework for evaluating processing-in-memory and heterogeneous systems across workload behavior, data movement, memory constraints, and hardware resources.
- Designing **Secure Scattered Memory (SSM)**, which explores secret-sharing-based protected-memory organization to reduce metadata traffic in secure memory systems.

Research Assistant, Columbia University *Jan 2019 – May 2019*

- Worked under Prof. Henning Schulzrinne in the Internet Real-Time Lab (IRT Lab).
- Designed and built an IoT system using machine-learning-based sound pattern classification.

Research Collaborator, Rolls-Royce, Indianapolis, IN *Sep 2017 – May 2018*

- Developed automotive attack/defense techniques for CAN bus communication.
- Implemented and validated techniques on a 2009 Honda Civic and a Tesla Model S.

Teaching Experience

Teaching Assistant, University of Notre Dame *2021 – Present*

- Teaching assistant for **Computer Architecture** and **Processing-in-Memory** courses at the undergraduate and graduate levels.
- Designed assignments and exam questions, developed autograders, graded coursework, and provided detailed technical feedback.
- Held office hours and mentored students on architecture and processing-in-memory coursework and projects.

Publications

Published and Accepted

[1] **H. Geng**, S. Wang, M. Li, T. S. Pereira, M. Niemier, X. Hu. *Towards Noise-Aware and Efficient Homomorphic Encryption on Crossbar Arrays. IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, 2026.

[2] **H. Geng**, X. Lu, Y. Che, Z. Tian, D. Cheng, X.-H. Sun, M. Niemier, X. Hu. *COSMOS: RL-Enhanced Locality-Aware Counter Cache Optimization for Secure Memory.* In *Proceedings of the 58th IEEE/ACM International Symposium on Microarchitecture (MICRO)*, 2025.

[3] **H. Geng**, J. Mo, D. Reis, J. Takeshita, T. Jung, B. Reagen, M. Niemier, X. Hu. *PPIMCE: An In-Memory Computing Fabric for Privacy-Preserving Computing. Journal of Computer Science and Technology (JCST)*, 2025.

[4] A. Mamdouh, **H. Geng**, M. Niemier, X. Hu, D. Reis. *Shared-PIM: Enabling Concurrent Computation and Data Flow for Faster Processing-in-DRAM. IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, 2025. *(Co-first author)*

[5] M. Li, **H. Geng**, M. Niemier, X. Hu. *Accelerating Polynomial Modular Multiplication with Crossbar-Based Compute-in-Memory.* In *Proceedings of the IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, 2023. *(Co-first author)*

[6] D. Reis, **H. Geng**, M. Niemier, X. Hu. *IMCRYPTO: An In-Memory Computing Fabric for AES Encryption and Decryption. IEEE Transactions on Very Large Scale Integration Systems (TVLSI)*, 2021.

[7] T. S. Pereira, R. Wang, I. Nwozo, **H. Geng**, X. Hu. *Codebook-Based Acceleration for Ternary Language Model Inference*. Accepted to appear in the *Proceedings of the IEEE/ACM International Symposium on Low Power Electronics and Design (ISLPED)*, 2026.

Selected Manuscripts Under Review

(First-author papers unless otherwise noted)

[8] **H. Geng**, Y. Che, D. Cheng, M. Niemier, X. Hu. *Secure Scattered Memory: High-Performance Memory Protection via Secret Sharing*. Submitted to *ISCA*, 2027.

[9] **H. Geng**, T. S. Pereira, X. Lu, X.-H. Sun, M. Niemier, X. Hu. *VIPER: A System-Level Performance Model for Exploring Processing-in-Memory Designs*. Submitted to *SIGMETRICS*, 2027.

Patents

[1] X. S. Hu, D. A. Reis, M. Niemier, **H. Geng**. *Circuits and Systems for Programmable In-Memory Encryption/Decryption Fabric Including RA/CAM Arrays for Parallel Processing of Operations*. U.S. Patent No. 12,470,364, issued Nov. 11, 2025.

Presentations

- *COSMOS: RL-Enhanced Locality-Aware Counter Cache Optimization for Secure Memory*, MICRO-58, Seoul, 2025.
- *Accelerating Polynomial Modular Multiplication with Crossbar-Based Compute-in-Memory*, ICCAD 2023, San Francisco, CA.
- Presented work on an in-memory computing accelerator for privacy-preserving computing, Techcon 2023, Austin, TX.

Honors & Awards

- Third Place Award, DAC PhD Forum, 63rd ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, 2026.

Academic Service

- Reviewer, *IEEE Transactions on Emerging Topics in Computing (TETC)*, 2024.
- Reviewer, *Design, Automation and Test in Europe Conference (DATE)*, 2025.
- Reviewer, *IEEE Transactions on Computers (TC)*, 2026.

Leadership & Outreach

Undergraduate Research Mentor, University of Notre Dame *2025 – Present*

- Mentor undergraduate researchers in in-memory computing, computer architecture, and emerging AI/LLM hardware.
- Guide students through experimental methodology, implementation, data analysis, and research development.
- Support student-led research toward independent publication and graduate-study preparation.

Professional Affiliations

- IEEE Student Member
- ACM Student Member

Professional Experience

Software Engineer Intern, OakGate Technology, Sacramento, CA *Jun 2019 – Sep 2019*

- Developed Python parsing scripts to enhance internal API functionality.
- Updated Bash scripts and C code for SSD device-discovery debugging.
- Created NVMe CMB SSD conformance-testing firmware in C.

References

Prof. Xiaobo Sharon Hu
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Prof. Michael Niemier
University of Notre Dame
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Prof. Brandon Reagen
New York University
bjr5@nyu.edu

Prof. Xian-He Sun
Illinois Institute of Technology
sun@illinoistech.edu